

Walk-forward performance (replay)

Fold × strategy × symbol from `backtest_executor.py --replay --include=trades` . **Expectancy** = mean PnL per trade; **PF** = $\text{sum}(\text{winning \$}) / |\text{sum}(\text{losing \$})|$; **Max DD** = peak-to-trough on cumulative PnL within that fold’s trade sequence; **ΣR** = $\text{sum of } (\text{trade PnL} / \text{initial_risk_dollars})$ where risk was recorded.

`metrics.json` · `metrics_insights.json`

All folds pooled: 18 trades · PnL **731.00** · expectancy **40.611** / trade · σ **94.118** · PF **2.69** · max consecutive losses **1** · max DD (sequential) **109.75**

Survivability read: high *max consecutive losses* inside a fold with negative expectancy usually means the edge is not stable across that window. Compare the *rollup* table (all folds merged per strategy×symbol) to per-fold rows — widening σ or falling PF in later folds is a simple walk-forward stress signal.

Per fold

fold	start	end	strategy	sym	n	PnL	WR%	E/trade	σ PnL	PF	maxDD	win	loss	max win str	max loss str	avg bars	ΣR
0	2026-04-26	2026-04-28	morning_range_reversion	MNQ	1	181.75	100.0%	181.750	0.000	—	0.00	1	0	1	0	13.0	0.95
1	2026-04-29	2026-05-01	morning_range_reversion	MNQ	2	57.00	50.0%	28.500	123.390	1.97	58.75	1	1	1	1	7.0	-0.18
2	2026-05-02	2026-05-04	morning_range_reversion	MNQ	1	-75.25	0.0%	-75.250	0.000	0.00	75.25	0	1	0	1	2.0	-1.08
3	2026-05-05	2026-05-07	morning_range_reversion	MNQ	2	152.00	100.0%	76.000	51.972	—	0.00	2	0	2	0	6.0	1.73
4	2026-05-08	2026-05-10	morning_range_reversion	MNQ	1	43.75	100.0%	43.750	0.000	—	0.00	1	0	1	0	2.0	0.82
5	2026-05-11	2026-05-13	morning_range_reversion	MNQ	3	197.25	100.0%	65.750	22.567	—	0.00	3	0	3	0	5.0	2.59
6	2026-05-14	2026-05-16	morning_range_reversion	MNQ	2	-47.50	50.0%	-23.750	121.622	0.57	109.75	1	1	1	1	1.0	-0.19
7	2026-05-17	2026-05-19	morning_range_reversion	MNQ	2	113.00	50.0%	56.500	228.749	2.07	105.25	1	1	1	1	1.0	-0.10
8	2026-05-20	2026-05-22	morning_range_reversion	MNQ	3	59.25	66.7%	19.750	90.276	1.71	83.75	2	1	2	1	4.7	0.69
9	2026-05-23	2026-05-26	morning_range_reversion	MNQ	1	49.75	100.0%	49.750	0.000	—	0.00	1	0	1	0	1.0	0.84

Rollup — all folds merged (per strategy × symbol)

strategy	sym	n	PnL	WR%	E/trade	σ PnL	PF	maxDD	max loss str	avg bars	ΣR
morning_range_reversion	MNQ	18	731.00	72.2%	40.611	94.118	2.69	109.75	1	4.3	6.08

Simulated account equity

Starting balance **\$2,000.00**. Every completed trade in this run is sorted by **exit time** and applied in sequence to one notional account (all legs, all folds). This shows combined PnL path stress; it does *not* model margin, cross-margin, or whether every leg would run on the same live account.

Equity path: final **2,731.00** · return **36.55%** · max drawdown **109.75 (4.19%** from peak) · closes in path **18**



Analysis — all legs pooled

All legs pooled (exit-time order, all folds)

Performance insights

Metric	Value	Note
Breakeven win rate (from avg win / avg loss)	49.2%	Actual WR 72.2%
Recovery factor (total PnL / sequential max DD)	6.661	Higher is better; uses fold-style cumulative PnL DD
Median bars held	W:5.0 / L:1.0	Win vs loss holding time
Avg R multiples	win:0.88 loss:-1.072	Where initial_risk_dollars recorded

Losses by exit reason

exit_reason	% of losses	detail
stop_loss	100.0%	losses 5 / events 5

Loss rate by side

side	Loss rate	detail
BUY	33.3%	3 losses / 9 trades
SELL	22.2%	2 losses / 9 trades

Loss mix by strategy leg

Populated when trades carry `_strategy` from the recap report (multi-strategy runs).

strategy	Losses	Context
morning_range_reversion	5	100.0% of all losses · 18 trades in leg

Loss count by entry weekday (ET)

Weekday	Losses	Context
Mon	1	20.0% of all losses; 3 entries this weekday
Tue	1	20.0% of all losses; 4 entries this weekday
Wed	1	20.0% of all losses; 4 entries this weekday
Thu	0	0.0% of all losses; 4 entries this weekday
Fri	2	40.0% of all losses; 3 entries this weekday

Linear association with losses (exploratory)

Pearson correlation between a 0/1 *loss indicator* and each factor. Values near 0 mean no obvious linear pattern in this sample; $|r|>0.15$ is a loose rule-of-thumb for “worth eyeballing” only.

Factor	r vs loss	Interpretation
bars_held	-0.3979	When bars_held is lower, losing trades are slightly more common (Pearson $r\approx-0.40$); treat as exploratory—causation not implied.
initial_risk_dollars	-0.1711	When initial_risk_dollars is lower, losing trades are slightly more common (Pearson $r\approx-0.17$); treat as exploratory—causation not implied.
entry_hour_et	0.2213	When entry_hour_et is higher, losing trades are slightly more common (Pearson $r\approx0.22$); treat as exploratory—causation not implied.

Analysis — per strategy × symbol (merged folds)

morning_range_reversion · MNQ (merged folds)

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